

Calvin Thrasher

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EDUCATION

Iowa State University – Ames, IA

Anticipated: May 2028

Bachelor of Engineering - Mechanical Engineering

PROFESSIONAL EXPERIENCE

Robotics Hardware Engineer | Stealth Startup | San Francisco, CA

July 2026 – August 2026

- Designed and fabricated a gripper-style UMI (Universal Manipulation Interface) device for manipulation data collection
- Designed a 200-DOF hand-tracking UMI device capturing full finger-joint kinematics with ground truth for manipulation data collection
- Managed production scaling of 200+ gripper-style UMI units, coordinating fabrication and quality control
- Designed a teleoperation rig optimized for backdrop randomization, scalability, and ergonomics
- Maintained, repaired, and configured YAM robotic arms to support ongoing data collection operations

Project Engineer Intern | Premier Tech | Johnston, IA

May 2026 – June 2026

- Modeled fabrication parts, load-bearing structure, spouting, dust pipe, and various other assemblies within Revit and SolidWorks for integration of Premier Tech and 3rd party equipment
- Produced 3D scans of various industrial sites for equipment integration

Project Engineer Intern | Keen Project Solutions | Ankeny, IA

May 2025 – August 2025

- Coordinated agricultural construction projects for Corteva Agriscience and AC Grain, contributing to on-time, budget-compliant delivery.
- Prepared project safety plans, tracked project milestones along with RFI/submittal statuses using Procore to maintain schedule and quality control.

LEADERSHIP & INVOLVEMENT

President | Cyber Robotics (Vex VRC)

April 2026 – Present

- Lead all executive operations, coordinate all internal and external communications, and drive strategic planning, meetings, outreach, and organization growth.

Mechanical Team Lead | Cyber Robotics (Vex VRC)

December 2024 – Present

- Lead mechanical sub team of 20+ members, overseeing CAD development, robot manufacturing, pneumatic/sensor integration, and assembly workflow.

Executive Competition Director | Cyber Robotics (Vex VRC)

August 2025 – April 2026

- Coordinated competition logistics, travel planning, university compliance, and event operations to ensure seamless team performance.

Projects

Cycloidal Drive Gearbox | Personal Project

April 2026 – Present

- Designed and fabricated a custom 20:1 cycloidal drive for NEMA 17 motors, targeting improved torque for compact robotic joints
- Engineering toward integration into a 6-axis robotic arm for object manipulation (in progress)

SKILLS

- **CAD/CAM** – SolidWorks, Inventor, Fusion 360, Revit, Cyclone 360; 3D-Printing (Bambu, Creality)
- **Programming** – C++, Python; Programming Algorithms (PID, Odometry, Sensor Macro Control)
- **Interpersonal** – Team Leadership, Technical Communication, Problem Solving, Attention to Detail

CERTIFICATIONS

- **Procore** – Project Management & Project Management Core Tools Certification
- **OSHA 10** – Occupational Safety and Health Certification